VIT Vidyalankar Institute of Technology
Accredited A+ by NAAC

Version 09/27-09

Consolidated Academic Administration Plan for the Course

Data Warehousing and Mining(DWM) Professional Core

Sem. V – Program INFT 2026-2027 –Odd Semester

Faculty -Dr. Vidya Chitre (Cluster Mentor) & Prof Shruti Agrawal

The academic resources available in VIT –

VMIS (ERP)	V-Refer and V-Live	VIT Library	VAC & MOOC Courses
Institute & Department Vision and Mission	Former IA question papers and solutions (prepared by faculty)	Former IA question papers solutions - hardcopy	Value Added Courses (VAC) are conducted throughout the semester & in the semester break - Enrol for the VACs
Program Educational Objectives (PEO)	MU end semester examination question papers and solutions (prepared by faculty)	MU end semester exam question paper & solutions - by faculty, hardcopy	
Program Specific Outcome (PSO)	Class notes and Digital Content for the subject (scanned / typed by faculty)	All text books, reference books, e -books mentioned in the syllabus & AAP	Online courses from NPTEL, Coursera etc. are pursued throughout the semester - Register for the course & get certified
Program Outcome (PO)	Comprehensive question bank, EQ, GQ, PPT, Class Test papers	Technical journals and magazines for reference	
Departmental Knowledge Map	Academic Administration Plan & Beyond Syllabus Activity report	VIT library has many resources e:g :- IEEE, Nimbus, xplore, EBSCO etc.	Watch former lectures captured in LMS at VIT

1.a Course Objectives (Write in detail – as per NBA guidelines)

Cognitive	What do you want students to know?	Students should understand how to analyze data and choose relevant models and algorithms for respective applications.
Affective	What do you want students to think / care about?	Students should think about designing data warehousing schemas and applying the right data mining algorithms effectively.
Behavioural	What do you want students to be able to do?	Students should cultivate a research interest in innovations in data mining.

Advice to Students:

Attend every class!!! Missing even one class can have a substantial effect on your ability to understand the course. Be prepared to think and concentrate, in the class and outside. I will try to make the class very interactive. Participate in the class discussions. Ask questions when you don't understand something. Keep up with the class readings. Start assignments and homework early. Meet me in office hour to discuss ideas, solutions or to check if, what you understand is correct.

The v-Refer Link

Creation of microsite or teams link – (Division wise)

General | Odd2627-TYINFT-C-DWM-SHA

Collaboration Policy:

We encourage discussion between students regarding the course material. However, no discussion of any sort is allowed with anyone on the assignment and homework for the class. If you find solution to some problems in a book or on the internet, you may use their idea for the solution; provided you acknowledge the source (name and page in the book or the website, if the idea is found on the internet). Even though you are allowed

to use ideas from another source, you must write the solution in your own words. If you are unsure whether or not certain kinds of collaboration is possible, please ask the teacher.

1.b Course Outcome (CO) Statements and Module-Wise Mapping (follow NBA guideline)

CO No.	Statements	Related Module/s
CO1	Design data warehouse schemas using dimensional (star, snowflake) and describe the ETL (Extract, Transform, Load) process involved in building and maintaining a data warehouse.	1
CO2	Recall and explain the fundamental concepts, architecture, and components of data warehouses, data marts, data lakes, and data mining systems.	2
CO3	Perform data preprocessing, OLAP operations, data exploration, visualization, and preprocessing techniques to prepare datasets for analysis.	3
CO4	Analyze classification and regression techniques such as decision tree, Naïve Bayes, simple linear regression, and multiple linear regression for predictive data mining.	4
CO5	Analyze clustering techniques such as K-Means, agglomerative clustering, and divisive clustering for grouping similar data objects.	5
CO6	Evaluate frequent pattern mining and association rule mining techniques such as Apriori, candidate generation, pattern growth, and vertical data format for decision-making.	6

1.c Course Outcome (CO) Statements mapping with Learning Outcome (LO)

	Mapped to Learning Outcomes
CO1	1.2, 1.3
CO2	1.1, 2.1, 2.2, 3.1
CO3	3.2, 3.3
CO4	4.1, 4.2
CO5	5.1, 5.2
CO6	6.1, 6.2, 6.3

1.d Mapping of COs with POs (As per new detailed Syllabus)

(List of POs is available in V-refer)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
	Knowledge A	Analysis	Design	Investigation	Modern Tools	Society	Environment &sustainability	Ethics T	Teamwork C	Communication	Project Mgt	Lifelong learning
CO 1	S	M	S	W	M	-	-	-	-	-	-	-
CO 2	S	M	-	-	W	-	-	-	-	-	-	-
CO 3	M	S	-	M	S	-	-	-	-	-	-	-
CO 4	M	S	-	M	M	-	-	-	-	-	-	-
CO 5	M	S	-	S	M	-	-	-	-	-	-	-
CO6	M	S	-	M	M	-	-	-	-	-	-	-

1.e Mapping of COs with PSOs (As per new detailed Syllabus)

	PSO 1	PSO 2	PSO 3	PSO 4
CO 1	S	M	W	-
CO 2	S	W	-	-
CO 3	M	S	M	-
CO 4	M	S	M	-
CO 5	M	S	M	-
CO6	M	S	S	-

1.f Teaching and Examination Scheme (As specified by the autonomous syllabus) for the Course

Verticals	BSC/ESC	Program Courses	Multidisciplinary Courses	Skill Courses	HSSM	Experiential Learning	Liberal Learning
Tick suitable category		√					

Subject Code	Subject Name	**Teaching Scheme**			Credits Assigned			
		Theory	Practical	Tutorial	Theory	TW/Practical	Tutorial	Total
PCIT10T	Data Warehousing & Mining	2	-	-	2	-	-	2
PCIT10P	Data Warehousing & Mining Lab	-	2	-	-	1	-	1

Subject Code	Subject Name					**Examination Scheme**			
		Theory			**Total (Theory)**	**Practical**			**Total (Practical)**
		ISA	MSE	ESE		ISA	ESE	ORAL	

PCIT10T	Data Warehousing& Mining	15	20	40	75				
PCIT10P	Data Warehousing& Mining Lab		-			25		25	50

Subject Code	Subject Name	**MSE-1***		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PCIT10T	Data Warehousing& Mining	Question 1	25% Module 3 25% Module 3	2
		Question 2	10% Module 2 40% Module 1	2

Subject Code	Subject Name	**MSE-2***		
		Q, No	Module wise % Distribution	Relevant to Bloom Taxonomy
PCIT10T	Data Warehousing& Mining	Question 1	25% Module 3 25% Module 3	3
		Question 2	25% Module 3 25% Module 4	3

Subject Code	Subject Name	**ESE#**		
		Q, No	Distribution	Relevant to Bloom Taxonomy
PCIT10T	Data Warehousing& Mining	Question 1	15% Module 1 10% Module 2	
		Question 2	15% Module 3 10% Module 4	
		Question 3	25% Module 5	
		Question 4	25% Module 6	

*** Recommended distribution: -** 30 Marks from Assignments, 40 marks based on assignments with slightly enhance difficulty /complex, 30 marks from thought provoking

Recommended distribution: - 30 Marks from Assignments, 40 marks based on assignments/MSE with slightly enhance difficulty /complex, 30 marks from thought provoking

1.g Faculty-Wise Distribution of all Lecture-Practical-Tutorial Hours for the Course

Divisions	Lecture (Hrs.)	Practical (Hrs.)				Tutorial (Hrs.)			
		Batch 1	Batch 2	Batch 3	Batch 4	Batch 1	Batch 2	Batch 3	Batch 4
A	VDC	VDC	SHA	VB	VB				
B	VDC	SHA	SHA	SHA	SHA				
C	SHA	SHA	SHA	SHA	SHA				

1.h Office Hours (Faculty will be available in office in this duration for solving students' query)

Division	Day	Time (at least 1 Hr. / Division)	Venue (Office Room No.)
A	Monday	4.00 pm to 5.00 PM	HOD cabin
B	Monday	3.00 pm to 4.00 PM	HOD cabin
C	Friday	4.00 pm to 5.00 PM	E105

Note: Office hours should be after academic hours of students.

2.a Syllabus: Module Wise Teaching Hours and % Weightage in autonomous syllabus Question Paper

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
1	Introduction to Data Warehouse and ETL Process	8	26%	25%	26%	PI 1.2.2, PI 2.1.2, PI 5.2.1
Learning Outcome-1.1	Explain the concepts, architecture, and components of a data warehouse and data mart.					
Learning Outcome-1.2	Design dimensional models using star, snowflake, fact constellation, factless fact, and aggregate fact tables.					
Learning Outcome-1.3	Describe the ETL process and perform basic OLAP operations for multidimensional analysis.					
2	Introduction to Data Lakes	3	10%	25%	10%	PI 1.2.2, PI 2.2.2, PI 5.3.4
Learning Outcome-2.1	Explain the architecture, characteristics, and functionalities of data lakes.					
Learning Outcome-2.2	Compare data lakes and data warehouses based on architecture, storage, and applications.					
3	Data Exploration and Data Preprocessing	5	16%	25%	16%	PI 2.3.2, PI 4.2.3, PI 5.2.3
Learning Outcome-3.1	Explain the KDD process and data mining system architecture.					

Module No.	Module Title and Brief Details	Teaching Hrs. for each module	% Weightage in autonomous syllabus Question Papers			Performance Indicator Mapping
			ISA	MSE	ESE	
Learning Outcome-3.2	Perform data exploration using statistical measures and visualization techniques.					
Learning Outcome-3.3	Apply data preprocessing techniques to prepare datasets for analysis.					
4	Classification	5	16%	25%	16%	PI 2.3.3, PI 4.3.3, PI 5.2.3
Learning Outcome-4.1	Explain the principles of classification and regression in data mining.					
Learning Outcome-4.2	Apply Decision Tree, Naïve Bayes, and regression techniques for predictive analysis.					
5	Clustering	4	14%		14%	PI 2.3.3, PI 4.3.3, PI 5.2.3
Learning Outcome-5.1	Explain the concept and applications of clustering techniques.					
Learning Outcome-5.2	Apply K-Means and hierarchical clustering algorithms to group similar data objects.					
6	Mining frequent patterns and associations	5	16%		16%	PI 2.3.3, PI 4.3.3, PI 5.2.4
Learning Outcome-6.1	Explain the concepts of frequent itemsets and association rule mining.					
Learning Outcome-6.2	Apply the Apriori algorithm to discover frequent patterns and generate association rules.					
Learning Outcome-6.3	Compare candidate generation, pattern growth, and vertical data format approaches for efficient frequent pattern mining.					
* Insert rows for more modules in the Course **Total**		**30**	**100%**	**100%**	**100%**	

Note: - As an attachment Annexure is required for assessment criteria of learning outcomes.

2.b Prerequisite Courses

No.	Semester	Name of the Course	Topic/s
1	IV	DBMS	DB design & fundamental concepts

2.c Relevance to Future Courses

No.	Semester	Name of the Course
1	VII	Deep learning , Big data Analytics
2	VII, VIII	Projects

2.d See :- Identify real life scenarios/examples which uses the knowledge of the subject ,(Discussion on how to prepare examples and case studies e.g. "Boeing Plane": C Programming Language – Intro to Computer Science – Harvard's CS50 (2018) – Bing video)

Real Life Scenario	Concept Used
1. Classifying medical images 2. Detecting tumours from MRI scans 3. Categorizing emails into spam or non-spam/ 4. Identifying abnormalities in X-rays or MRI scans.	Classification Algorithm
1. Weather/ Stock Market Prediction 2. Predicting patient recovery time	Regression Algorithm
1. Customer or patient segmentation 2. To identify anomalous network traffic behavior 3. detect cybersecurity threats. 4. Credit card fraud detection.	Clustering Algorithm
1. Market Basket Analysis 2. Identifying co-occurring symptoms in medical records 3. Analysing DNA sequences to discover genetic markers associated with specific diseases 4. Examining social media interactions to identify influential users	Association Mining Algorithm

3 Past Results – Division-Wise

Details	Target – DEC 2026	MAY 2025	DEC 2024	MAY 2024
Course Passing % – Average of 2 Divisions	99	100	94	100
Marks Obtained by Course Topper (mark/100)	75	75	NA	70

	Division A		Division B		Division C	
Year	Initials of Teacher	% Result	Initials of Teacher	% Result	Initials of Teacher	% Result
May 2025	VDC	100	VDC	99	VDC	99
May 2024	VDC	100	SHA	100	-	-
May 2023	NA	NA	NA	NA	-	-

4 All the Learning Resources – Books and E-Resources

4.a List of Textbooks (T – Symbol for Textbooks) to be Referred by Students

Sr. No	Textbook Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Data Mining Concepts and Techniques	Han, Kamber	Elsevier	3rd Edition	3,4,5	Yes

2	Data Warehousing: Fundamentals for IT Professionals.	Paulraj Ponniah,	Wiley India	3rd Edition	1	No
3	Data Lake Architecture	Bill Inmon,	Technics Publication	1st Edition	2	No
4	The Data Warehouse Toolkit	Margy Ross and Ralph Kimball,	Willey	3rd Edition	1	Yes

4.b List of Reference Books (R – Symbol for Reference Books) to be Referred by Students

Sr. No	Reference Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Building the Data Warehouse	W. H. Inmon	Wiley Computer Publishing	3rd edition,	1	Yes
2	Data Mining	Ian H. Witten, Eibe Frank and Mark A. Hall	Morgan kaufmann publisher	3rd edition	3,4,5	No
3	Introduction to Data Mining	Pang-Ning Tan, Vipin Kumar, Michael Steinbach	Pearson Education	2nd edition	3,4,5	Yes

4.c List of E - Books (E – Symbol for E-Books) to be Referred by Students

Sr. No	E- Book Titles	Author/s	Publisher	Edition	Module Nos.	Available in our Library
1	Data Warehousing: Fundamentals for ITProfessionals.	Paulra jPonniah,	Wiley India	3nd edition	1,2	NO
2	Data Mining Concepts and Techniques	Han, Kamber	Elsevier	3nd edition	3,4,5	NO

4.d Reading latest / top rated research papers (at least 5 papers)

Name of Paper	Name of Authors (Background)	Published in		Problem Statement	Available in our Library
		Date	Journal		
More Than Beautiful: Exploring Design Features, Practical Perspectives, and Implications of Artistic Data Visualization	Xingyu Lan et al. Professor ,Fudan University	Feb 7, 2025	arxiv.org.	How can we develop a structured evaluation framework that captures both the aesthetic value and communicative effectiveness of artistic data visualizations, while ensuring interpretability and data fidelity across diverse user groups?	NO [Online Available on https://arxiv.org/pdf/2502.04940v1]
Hierarchical Contrastive Learning for Fine-Grained Image Classification	Cheng Jin, Department of Computer Science and Engineering, The Hong Kong University of Science and Technology, Hong Kong, SAR, China	April 2025	IEEE Transactions on Medical Imaging, vol. 44, no. 4, pp. 1796-1808, April 2025, doi: 10.1109/TMI.2024.3520602	How can hierarchical relationships between classes be leveraged to improve feature discrimination in fine-grained image classification where subtle differences cause misclassification?	Yes
Design and Performance Evaluation of Efficient Clustering Algorithms for Big Data Applications"	Ping Dai , Anhui Institute of Information TechnologyWuhu, China	Feb 2025	*Science Direct - Applied Mathematics and Nonlinear Sciences*, 10(1), 1-12. DOI: https://doi.org/10.2478/amns-2025-0056	How can clustering algorithms be improved to efficiently handle high-dimensional big data while maintaining accuracy and overcoming challenges like attribute influence and initialization sensitivity?	No
Workload-Aware Incremental Reclustering in Cloud Data Warehouses	Yipeng Liu, Renfei Zhou, Jiaqi Yan, Haunchen Zhang (Database Systems Researchers)	February 2026	arXiv / ACM Preprint	How can cloud data warehouses incrementally reorganize data to improve query performance while minimizing reclustering overhead and storage costs?	No
DTMiner: A Data-Centric System for Efficient	Yinbo Hou, Hao Qi, Ligang He, Yu Zhang, Hui Yu et al.	January 2026	PPoPP 2026 (University	How can temporal motif mining on massive dynamic graphs be accelerated using a data-centric architecture while	No

Temporal Motif Mining	(Researchers from leading universities)		of Warwick Repository)	maintaining scalability and high performance?	

4.e **Based on research paper an identify the current Problem statement**

Problem Statement			Used in			
	Quiz	Assignment	*Mini Project*	Poster Presentation	Test	Any Other
1						Paper Review
2						Paper Review
3						Paper Review
4						Paper Review
5						Paper Review

4.f **Identify Companies / Industries which use the knowledge of the subject and thus may provide Internships and final Placements**

Name of the Company	To be / Contacted for		
	Student Internship	Student Final Placement	Faculty Internship
1. Oracle Data Mining 2. IBM 3. CDAC 4. CreatED	Student Internship		

4.g **Identify suitable relevant TOP Guest Speakers from Industry, Example: - (CS50 Lecture by Mark Zuckerberg - 7 December 2005 - YouTube)**

Name of the Identified Guest Speaker	Designation	Name of the Company
Mr. Hemant Tendolkar	Enterprise Architect, AWS Business Unit	TCS

4.h **Identify relevant technical competitions to participate [Competitions -Paper Presentations, Projects, Hackathons, IVs etc..]**

Name of the Relevant Technical Competition Identified to participate	Organized by	Date of the Event
Smart India Hackathon (SIH) 2026 (AI, Big Data & Analytics Problem Statements)	Ministry of Education & AICTE	Aug–Dec 2026
Kaggle Data Science & Machine Learning Competitions	Kaggle	Throughout 2026–27
IEEE International Conference on Data Mining (ICDM 2026) Student Research Forum / Poster Competition	Institute of Electrical and Electronics Engineers	Dec 2026
ACM Student Research Competition (SRC) (Data Mining / AI Track)	Association for Computing Machinery	2026 (Conference-wise)

4.i **Identify faculty in TOP schools / Universities who are teaching same / similar subject and develop rapport e.g. Exchange Lecture Material (Assignments / Tests / Project etc..), Joint Paper Publication**

University	Name of the Course	Name of Faculty	Type of Collaboration		
			Exchange of Lecture Material	Joint Publication/ Research	Other
IIT Kharagpur	Data Mining	Prof. Pabitra Mitra	Lecture notes		
NITT Tiruchirappalli	Data Warehousing and Data Mining	Dr. E Sivasankar	Chapter 1 PPT		

4.j **Module Best Available in – Title of the best resource [from *4.a* to *4.d* in this AAP] and other details as necessary**

Module No.	Title of the Module	Textbook	Mention the Tile					
			Reference Book	E-books	Journal	E-Journal	Available in our Library	V-refer
1	Introduction to Data Warehouse and ETL Process	Datawarehousing and OLAP operation	A Review: Analysis on Data Warehousing and Data Mining					Y
2	Introduction to Data Lakes		IEEE Transactions on Knowledge and Data Engineering		Introduction to Data Lakes	https://www.altexsoft.com/blog/data-lake-architecture/		Y
3	Data Exploration and Data Preprocessing		Data Mining and Knowledge Discovery Springer			https://youtu.be/NSxEiohAH5o		Y
4	Classification		International Journal of Data Mining, Modelling and Management. Inder Science			https://archive.nptel.ac.in/courses/		Y

5	Clustering	https://www.youtube.com/watch?v=ZTP6b3LaVGg	Statistical Analysis and Data Mining Wiley					
6	Mining frequent patterns and associations	Association Mining	Frequent Pattern Mining Algorithms for Finding Associated Science Direct					Y

4.k **Referred to any top-rated university in that subject for content**

University	Name of the Course	Name of Faculty	Date of Delivery of the Course	Remarks
Technische Universität Braunschweig	Data Warehousing and Data Mining Techniques	Wolf-Tilo Balke	Summer Semester 2026	Covers Data Warehouse Design, Storage, OLAP, Knowledge Discovery, Data Mining Procedures, and introduction to Deep Learning. (Technische Universität Braunschweig)
The University of Hong Kong	COMP7103C – Data Mining	Chao Huang	Semester 1 (2024–25)	Covers Data Mining Architecture, OLAP, Data Preprocessing, Association Rule Mining, Classification, Clustering, Web, Spatial & Temporal Data Mining. (msccs.cs.hku.hk)
IIT Kharagpur	Data Mining	Prof. Pabitra Mitra	Feb 2021	https://youtu.be/PiGtQK7_fg4
Bits Pilani	Data Science and Engineering	Abhishek Kumar, Mishra	online	Explore the concept of Data Warehousing and Machine learning

4.l **Faculty received any certification related to this subject. List of Certifications Identified / Done**

Course	Certifying Agency	No. of Hours	Level of the Course		Certification		Remarks
			Introductory	Advance Skill Development	Done on	Proposed to be on	
Machine Learning A_Z	Udemy	43	--	✓	June 2025		-

4.m **Completed subject wise/cluster wise training with cluster mentor. List of relevant Refresher Course Identified / Done**

Course	Certifying Agency (As suggested by DAB/Cluster Mentor/Industry/ University other than MU)	Certification		Remarks
		Done on	Proposed to be on	
Pedagogy	The Complete Train the Trainer Bootcamp (Part 1): All Levels	2020	--	Successfully completed
PBL	100 Days of Code: The Complete Python Pro Bootcamp	2024	--	Successfully completed
	The Data Science Course: Complete Data Science Bootcamp 2024	2024	--	Successfully completed
	Complete Data Science, Machine Learning, DL, NLP Bootcamp	2025		Successfully completed
	Cluster Meeting	2026	--	Successfully completed
Sub. Content Training	Deep Learning and NLP: Seq2Seq Model Theory + ChatGPT Prizes	2024	--	Successfully completed

4.n Best Practices Identified and adopted

No.	Item	Best Practices Identified		
		IIT	Harvard	MIT
1	Microsite	–	–	–
2	Video Lectures	NPTEL Courses	Harvard Online Learning	MIT OpenCourseWare
3	Assignments	NPTEL Courses (Assignments)	–	–
4	Mini Project	NPTEL Project-Based Courses	Harvard Graduate School of Education – Project-Based Assessment	–
5	Assessment Metric	–	–	–
6	Quizzes	NPTEL (Assignments & Quizzes)	Harvard Online Learning	–
7	Labs/ Practical (PBL)	–	–	–
8	Tests	–	–	–
9	Peer Assessment	NPTEL SWAYAM Discussion & Peer Learning	Harvard Medical School – Peer Assessment in Teaching	–
10	Any Other	–	–	–

4.o Web Links for Online Notes/YouTube/VIT Digital Content/VIT Lecture Capture/NPTEL Videos

Students can view lectures by VIT professors, captured through LMS 'Lecture Capture' in VIT campus for previous years.

No.	Websites / Links	Module Nos.
1	https://youtu.be/PiGtQK7_fg4	All
2	https://nptel.ac.in/courses/106105174	2 to 6
3	https://www.coursera.org/specializations/data-mining	3 to 6

4.p Recommended MOOC Courses like Coursera / NPTEL / MIT-OCW / edX/VAC etc.

Sr. No.	MOOC Course Link	Course conducted by – Person / University / Institute / Industry	Course Duration	Certificate (Y / N)
1	https://www.coursera.org/learn/data-warehousing-business-intelligence	Data Warehousing and Business Intelligence – University of California, Irvine	Approx. 16 Hours (Self-paced)	Y
2	https://nptel.ac.in/courses/106106853	**NPTEL- Data Mining for Decision Making, IIT Mandi** Prof. Varun Dutt	4 weks	Y

5 Consolidated Course Lesson Plan

	From (date/month/year)	From (date/month/year)	Total Number of Weeks
Semester Duration	06/07/2026	23/10/2026	16

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Recommended Prior Viewing / Reading	
						Lecture No. (on LMS)	Chapter No./ Books/ Web Site
1	1	1	Chapter 1: Introduction to Data Warehouse and ETL Process Introduction to Data Warehouse and Data Mart, Data Warehouse architecture.		CO1		Fundamentals PaulrajPonnianh Chapter 15 page no 373– 405
	2	1	Data Warehouse vs Data Marts Information Package Diagram		CO1		Data Mining Concepts and Techniques, Han, Kamberpage no 135-164
2	3	1	Data Warehouse Schemas; Star Schema, Snowflake Schema, Fact Constellation Schema.		CO1	yes	Fundamental PaulrajPonnianh Chapter 15 page no 373– 405
			Factless Fact Table, Update to the dimension tables.			yes	Data MiningConcepts and Techniques Han,

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Recommended Prior Viewing / Reading	
						Lecture No. (on LMS)	Chapter No./ Books/ Web Site
	4	1			CO1		Kamber, page no 135-164
3	5	1	OLAP operations		CO1	Yes	Data Mining Concepts and Techniques,Han, Kamberpage no - 327-434
	6	1	Rollup, drill down, Slice, Dice, pivot		CO1	Yes	Data Mining Concepts and Techniques, Han, Kamber page no - 327-434
4	7	1	Basic steps of the ETL process, different extraction methods, transformations		CO1	Yes	Data MiningConcepts and Techniques, Han, Kamberpage no - 248-265
	8	1	Different loading techniques		CO1	Yes	Data MiningConcepts and Techniques, Han, Kamberpage no - 248-265
5	9	2	Introduction to Data Lakes Definition, key attributes of the data lake		CO2	Yes	Data Lake ArchitectureBill Inmon, Page no 30-37
	10	2	challenges, functionalities, architecture, Curating data lakes		CO2	Yes	Data Lake Architecture Bill Inmon, Page no 47-63

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Recommended Prior Viewing / Reading	
						Lecture No. (on LMS)	Chapter No./ Books/ Web Site
6	11	2	Data Lake vs. Data Warehouse		CO2	Yes	Data Lake Architecture Bill Inmon, Page no 63-77
	12	3	**Data Exploration and Data Preprocessing** The KDD process, Data mining system architecture, Data Exploration: Types of Attributes		CO3	Yes	Data Mining Concepts and Techniques, Han, Kamberpage no 83-117
7	13	3	Statistical Description of Data		CO3	Yes	Data MiningConcepts and Techniques, Han, Kamberpage no 83-117
	14	3	Data Visualization: box plots, line & bar charts, and scatter plots.		CO3	Yes	Data Mining Concepts and Techniques, Han, Kamber page no - 327-434,
8	15	3	Data Preprocessing: Descriptive data		CO3	Yes	Data Mining Concepts and Technique Han, Kamber page no - 327-434
	16	3	summarization, Cleaning, Integration &		CO3	Yes	Data Mining Concepts and TechniqueHan,Kamber page no - 248-265
9	17	3	transformation, Data reduction.		CO3	Yes	Data Mining Concepts and Techniques, Han, Kamber page no - 248-265
	18	4	Introduction to data mining techniques, Classification:		CO4	Yes	Data Mining Concepts and Techniques, Han, Kamberpage no 83-117

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Recommended Prior Viewing / Reading	
						Lecture No. (on LMS)	Chapter No./ Books/ Web Site
10	19	4	Decision Tree Induction		CO4	Yes	Data MiningConcepts and Techniques, Han, Kamberpage no 83-117
	20	4	Naïve Bayesian Classification.		CO4	Yes	Data MiningConcepts and Techniques, Han, Kamberpage no 83-117
11	21	4	Simple Regression		CO4	Yes	Data MiningConcepts and Techniques, Han, Kamberpage no 83-117
	22	4	Multiple Regression		CO4	Yes	Data MiningConcepts and Techniques, Han, Kamberpage no 83-117
12	23	5	Clustering: Partition based:		CO5	Yes	Data Mining Concepts and Techniques, Han, Kamberpage no 444-487
	24	5	K-means Clustering		CO5	Yes	Data MiningConcepts and Techniques, Han, Kamber
13	25	5	Hierarchical Methods (Agglomerative, Divisive)		CO5	Yes	Data MiningConcepts and Techniques, Han, Kamber
	26	6	(Agglomerative, Divisive)		CO5	Yes	Data MiningConcepts and Techniques, Han, Kamber
14	27	6	Basic Concepts: Market Basket Analysis, Frequent Itemset, Closed Itemset, and Association Rules; The		CO6	Yes	Data MiningConcepts and Techniques, Han, Kamber

Week	Lecture no.	Module No.	Lecture Topics / MSE / BSA planned to be covered	Actual date of Completion (Handwritten)	COs Mapped	Recommended Prior Viewing / Reading	
						Lecture No. (on LMS)	Chapter No./ Books/ Web Site
			Apriori Algorithm: Finding Frequent Itemset Using Candidate Generation, Generating Association Rules from requent Itemset,				
	28	6	Improving the Efficiency of Apriori, A pattern growth approach for mining		CO6	Yes	Data MiningConcepts and Techniques, Han, Kamber
15	29	6	Frequent Itemset. Mining Methods:		CO6	Yes	Data MiningConcepts and Techniques, Han, Kamber
	30	6	Frequent Itemset, Mining		CO6	Yes	Data MiningConcepts and Techniques, Han, Kamber
			Revision		CO6	Yes	

6 Rubric for Grading and Marking of Term Work (inform students at the beginning of semester)

- Activity/ies should be designed as per reference of credit structure.
- If the subject is of 2 credit, activity/ assignment should be design for 2 hours with appropriate complexity and engaging time.

Theory (ISA=15 Marks)												Total
Class Participation	Assignment -1	Assignment -2	Assignment -3	Assignment -4	Assignment -5	Assignment -6	Assignment -7	Assignment -8	Assignment -9	Assignment -10	Assignment -11	
02	01	01	01	01	01	01	01	01	01	01	03	15

Practical (ISA=25)	Total

Class Participation	Practical -1	Practical -2	Practical -3	Practical -4	Practical -5	Practical -6	Practical -7	Practical -8	Practical -9	Practical -10	Practical -11 & 12	
05	01	01	01	01	01	01	01	01	01	05	05	25

Class Participation	MSE-1	MSE-2	ESE	Total
15	10	10	40	75

7 Assignments / Tutorials Details

Assignment/ Tutorial No.	Title of the Assignments / Tutorials	CO Map	Mapping Bloom Taxonomy Level	Assignment/ Tutorials given to Students on	Assignments to be submitted back on
1	MOOC Course Enrolment & Registration	CO6	L2	14/07/2026	21/07/2026
2	Data Warehouse Poster & Case Study / OLTP vs DW Analysis	CO1, CO2	L4	22/07/2026	28/07/2026
3	IEEE Paper Summary & Star vs Snowflake Schema Comparison	CO2, CO6	L5	29/07/2026	04/08/2026
4	Flip Classroom & Regression Model	CO4	L3	05/08/2026	11/08/2026
5	Data Preprocessing & Exploratory Data Analysis	CO3	L3	12/08/2026	18/08/2026
6	Open Book Test	CO1–CO5	L3	19/08/2026	25/08/2026
7	Decision Tree, Naïve Bayes & Information Gain	CO4	L4	27/08/2026	01/09/2026
8	K-Means Clustering Activity	CO4	L3	02/09/2026	08/09/2026
9	Apriori Algorithm & Association Rule Mining	CO5	L4	09/09/2026	15/09/2026
10	Take Home Test	CO1–CO5	L4	23/09/2026	30/09/2026
11	MOOC Course Certificate Submission	CO6	L2	01/10/2026	01/10/2026

Analysis of Assignment / Tutorial Questions and Related Resources

Assignment / Tutorial No.	Week No.	Type* (√)			Module No.	Based on #			Question Type (√)	
		OT	CS	DTP		Textbook	Reference Book	Other Learning Resource	Real Life Assignments	Thought Provoking
1	Week 2	✓			All	✓		✓		✓
2	Week 3		✓	✓	1	✓	✓	✓	✓	✓
3	Week 4		✓		1	✓	✓	✓	✓	✓
4	Week 5			✓	4	✓	✓	✓	✓	
5	Week 6			✓	3	✓	✓	✓		✓
6	Week 7	✓			1–5	✓	✓			✓
7	Week 8		✓	✓	4	✓	✓	✓	✓	✓
8	Week 9			✓	5	✓	✓	✓		✓
9	Week 10			✓	5	✓	✓	✓	✓	✓
10	Week 12	✓			1–5	✓	✓			✓
11	Week 13	✓			All		✓	✓		

* Tick (√) the Type of the Assignment: Online Tools (OT); Collaborative Assignments (CS); Design /Thought provoking (DTP)
Write number for textbook, reference book, other learning resource from this AAP – *from Points 4.a to 4.d*

8 In Semester Assessment (ISE) / Other Class Test / Open Book Test (OBT)/Take Home Test (THT) Details

Tests	Test Dates	Module No.	CO Map	MSE Question Paper Pattern	Policy
ISE		MSE 1- 1,2 MSE 2- 3,4	CO1,CO2, CO3, CO4	Q.1 -1 questions out of 2 -10 M Q.2 -1 questions out of 2 -10 M	
Pop Quiz	Week 9	5	CO5, CO6	10 question of 1 Marks each	
Open Book Test	Week 7	3	CO5	two question of 10 Marks each	
Take Home Test	Week 12	6	CO6	Two question of 10 Marks each	
Class tests / prelims					
Any other test/exams					

*** Failures of IA test (IA1+IA2) shall appear for IA test in the next semester. There is no provision for re-test in the same semester.**

9. Practical Activities

Practical No.	Module No.	Title of the **Experiments**	Type of Experiment		Topics to be highlighted	CO Map
			PBL	Newly Added		
1	1	Build a Data Warehouse/Data Mart and design Star, Snowflake & Fact Constellation Schemas			Data Warehouse Schema Design	CO1
2	1	Perform OLAP Operations on the Dimensional Model			OLAP Operations	CO1
3	3	Implement Data Visualization Techniques on a Given Dataset			Data Visualization	CO3
4	4	Implement Simple and Multiple Linear Regression using Python			Regression Analysis	CO4
5	4	Perform Attribute Relevance Analysis and Decision Tree Classification			Decision Tree Classification	CO4
6	4	Implement Naïve Bayes Classification and Evaluate the Model			Naïve Bayes Classification	CO4
7	5	Implement K-Means Clustering using Python/Weka/RapidMiner			K-Means Clustering	CO4
8	5	Implement Agglomerative Clustering using Python			Hierarchical Clustering	CO4
9	6	Implement Apriori Algorithm for Association Rule Mining			Association Rule Mining	CO4
10	1–6	Mini Project: Data Preprocessing, Classification, Clustering & Association Analysis using WEKA/R			Integrated Data Mining Project	CO6
11	3 (PBL)	Build a Predictive Model for AQI or House Price Prediction	✓		Regression-Based Predictive Analytics	CO5
12	4 (PBL)	Customer Segmentation using K-Means Clustering for Retail Analytics	✓		Customer Segmentation & Clustering	CO5

10 Uncovering syllabus with different Activities.

No.	Type of the Activity	Activities	Number of beneficiaries	Other Details – guest profile, feedback, mark sheet, report
1	**Experiential learning/Interaction with Outside World**	1- Guest Lectures by Industry Expert	All students of TEINFT	TCS Data Mining
		2- Workshops	Industrial Visit	--
		3- Mini Project	All students of TEINFT	Based on Mind Map
		4- Industrial Visit	-- Not planned	--
		5- Any other activity	All students of TEINFT	Student will enroll for value added course
2	**Collaborative & Group Activity**	6- Poster Presentation	All students of TEINFT	Poster Presentation
		7- Minute Papers	All students of TEINFT	Minute Papers
		8- Students Seminars	All students of TEINFT	Students will present seminar on BI Tool
		9- Students Debates	Not planned	
		10- Panel Discussion / Mock GD	Not planned	
		11- Mock Interview	Not planned	
		12- Any other activity	Class Tests / Weekly Tests Lecture Capture Usage	Yes
3	**Co-Curricular Activity**	13- Informative videos (NPTEL/YouTube /TEDx/ MIT OW/edX)		Yes NPTEL Video Links for Topic: Data Mining & Knowledge Discovery https://www.youtube.com/watch?v=m5c27rQtD2E https://www.youtube.com/watch?v=0hnqxIsXcy4
		14- Lecture Capture Usage	Pop Quiz	**Yes**

		15- Any other activity	Mobile App Based Quiz	--
4	**Tests & Assessments**	16- Class Tests/ Weekly Tests	Open Book and Take Home Test	**Yes**
		17- Pop Quiz	All TEINFT students	
		18- Mobile App Based Quiz	Not planned	
		19- Open Book Test	All TEINFT students	
		20- Take Home Test	Not Planned	
		21- Any other activity	All TEINFT students	

11 AAP

No.	Programme	Course	Uploaded on V-refer	Date
1	INFT	DWM	Yes	9/07/2026

12 Lecture Guide

No.	Programme	Course	Uploaded on V-refer	Date
1	INFT	DWM	Yes	9/07/2026

*** Do not delete any activity. Give details for planned events. Write 'NA' for activity Not Planned.**

Consolidated Academic Administration Plan Prepared by (mention all theory teaching faculty names with signature)

Please write below your name and sign with date of the external cluster mentor meeting

Dr.Vidya Chitre Faculty 1	Prof. Shruti Agrwal Faculty 2	Dr.Varsha Bhosale Faculty 3

Mr.Harsh Batia External Industry Mentor	Dr. H.K. Chavan External Academic Mentor	Dr.Vidya Chitre VIT Cluster Mentor	Dr.Vidya Chitre Program HOD

Annexure:

Assessment Criteria of Learning Outcomes:

Learning Outcomes: The Learner will:	**Assessment Criteria:** The Learner can:	**Evaluated under ISA/MSE/ESE/LAB**
Explain the concepts, architecture, and components of a Data Warehouse and Data Mart.	Describe the architecture and components of a Data Warehouse and distinguish it from operational databases.	ISA, MSE, ESE
Design dimensional models using Star, Snowflake, Fact Constellation, Factless Fact, and Aggregate Fact tables.	Design appropriate dimensional models for a given business scenario.	ISA, ESE, LAB
Explain the ETL process and perform OLAP operations.	Demonstrate ETL workflow and perform Roll-up, Drill-down, Slice, Dice, and Pivot operations.	ISA, LAB
Explain the architecture and characteristics of Data Lakes.	Describe the architecture, components, and applications of Data Lakes.	ISA, MSE, ESE
Compare Data Lakes and Data Warehouses.	Differentiate Data Lakes and Data Warehouses based on architecture, storage, processing, and use cases.	ISA, ESE
Explain the KDD process and Data Mining system architecture.	Describe the Knowledge Discovery Process and Data Mining architecture.	ISA, MSE
Perform data exploration using statistical measures and visualization techniques.	Analyze datasets using descriptive statistics and visualization tools.	LAB, ISA
Apply data preprocessing techniques.	Perform data cleaning, transformation, normalization, and feature selection on datasets.	LAB, ISA
Explain the principles of Classification and Regression.	Describe classification and regression techniques and identify suitable algorithms for a given problem.	ISA, MSE, ESE
Apply Decision Tree, Naïve Bayes, and Regression techniques.	Build and evaluate predictive models using classification and regression algorithms.	LAB, ISA, ESE
Explain and apply Clustering techniques.	Implement K-Means and Hierarchical Clustering and interpret clustering results.	LAB, ISA
Explain and apply Frequent Pattern Mining and Association Rule Mining.	Implement the Apriori algorithm, generate association rules, and compare frequent pattern mining approaches.	LAB, ISA, ESE